

Low-density lipoprotein and high-density lipoprotein particle subclasses predict coronary events and are favorably changed by gemfibrozil therapy i...

BACKGROUND: Changes in conventional lipid risk factors with gemfibrozil treatment only partially explain the reductions in coronary heart disease (CHD) events experienced by men in the Veterans Affairs High-Density Lipoprotein Intervention Trial (VA-HIT). We examined whether measurement of low-density lipoprotein (LDL) and high-density lipoprotein (HDL) particle subclasses provides additional information relative to CHD risk reduction. **METHODS AND RESULTS:** This is a prospective nested case-control study of 364 men with a new CHD event (nonfatal myocardial infarction or cardiac death) during a 5.1-year (median) follow-up and 697 age-matched controls. Nuclear magnetic resonance (NMR) spectroscopy was used to quantify levels of LDL and HDL particle subclasses and mean particle sizes in plasma obtained at baseline and after 7 months of treatment with gemfibrozil or placebo. Odds ratios for a 1-SD increment of each lipoprotein variable were calculated with adjusted logistic regression models. Gemfibrozil treatment increased LDL size and lowered numbers of LDL particles (-5%) while raising numbers of HDL particles (10%) and small HDL subclass particles (21%). Concentrations of these LDL and HDL particles achieved with gemfibrozil were significant, independent predictors of new CHD events. For total LDL and HDL particles, odds ratios predicting CHD benefit were 1.28 (95% CI, 1.12 to 1.47) and 0.71 (95% CI, 0.61 to 0.81), respectively. Mean LDL and HDL particle sizes were not associated with CHD events. **CONCLUSIONS:** The effects of gemfibrozil on NMR-measured LDL and HDL particle subclasses, which are not reflected by conventional lipoprotein cholesterol measures, help to explain the demonstrated benefit of this therapy in patients with low HDL cholesterol.